

In class on Wednesday we saw the following context-free grammar. The grammar is on the left; the explanation for the last non-terminal is on the right. In order to define $L(S)$, let's first figure out what $L(A)$ and $L(B)$ are.

$$\begin{aligned}
 S &\rightarrow A \mid B \\
 A &\rightarrow 0A \mid 0C \\
 B &\rightarrow B1 \mid C1 \\
 C &\rightarrow \varepsilon \mid 0C1
 \end{aligned}
 \qquad
 \{0^n 1^n \mid n \geq 0\}$$

1. What is $L(A)$?
2. What is $L(B)$?
3. What is $L(S)$?

Give context-free grammars for each of the following languages over the alphabet $\Sigma = \{0, 1\}$. For each grammar, describe the language for each non-terminal, either in English or using mathematical notation, as in the examples above. We probably won't finish all of these during the problem session.

4. All palindromes in Σ^*
5. All palindromes in Σ^* that contain an even number of 1s
6. All palindromes in Σ^* that end with 1
7. All palindromes in Σ^* whose length is divisible by 3
8. All palindromes in Σ^* that do not contain the substring 00

Harder problems to work on later:

6. $\{0^{2n}1^n \mid n \geq 0\}$

7. $\{0^m1^n \mid m \neq 2n\}$

[Hint: If $m \neq 2n$, then either $m < 2n$ or $m > 2n$. Extend the previous grammar, but pay attention to parity. This language contains the string 01 .]

8. $\{0, 1\}^* \setminus \{0^{2n}1^n \mid n \geq 0\}$

[Hint: Extend the previous grammar. What's missing?]

9. $\{w \in \{0, 1\}^* \mid \#(0, w) = 2 \cdot \#(1, w)\}$ — Binary strings where the number of 0s is exactly twice the number of 1s.

10. $\{0, 1\}^ \setminus \{ww \mid w \in \{0, 1\}^*\}$.

[Anti-hint: The language $\{ww \mid w \in \{0, 1\}^*\}$ is **not** context-free. Thus, the complement of a context-free language is not necessarily context-free!]